

Primary sources & further reading

Part 01 —  Machine Learning Foundations · 01-ml-foundations/papers/PAPERS.md · 01-ml-foundations/resources/RESOURCES.md



Papers & Primary Sources — ML Foundations

Run `bash ../_tools/fetch_papers.sh 01-ml-foundations` from the repo root to download the open-access PDFs into this folder. Items without a direct PDF link are noted.

Read these two first

#	Paper	Why it matters	Link
1	A Few Useful Things to Know About Machine Learning — Pedro Domingos (2012)	The single best 9-page summary of ML practice ever written. "More data beats a cleverer algorithm." Read it twice.	PDF
2	Hidden Technical Debt in Machine Learning Systems — Sculley et al., NeurIPS 2015	Why the model is 5% of a real ML system. Introduces CACE: <i>Changing Anything Changes Everything</i> . Foundational for Module 10 and 16.	PDF

Core theory & method

Paper	Year	One-line takeaway	Link
Random Forests — Breiman	2001	Bagging + random feature subsets kills variance	PDF
Greedy Function Approximation: A Gradient Boosting Machine — Friedman	2001	The original gradient boosting derivation	PDF
XGBoost: A Scalable Tree Boosting System	2016	The engineering that made boosting the tabular default	arXiv:1603.02754
Statistical Modeling: The Two Cultures — Breiman	2001	Prediction vs. inference; the intellectual split that still shapes the field	PDF
Reconciling Modern Machine Learning and the Bias-Variance Trade-off — Belkin et al.	2018	Double descent; the textbook U-curve is incomplete	arXiv:1812.11118
Do we need hundreds of classifiers? — Fernández-Delgado et al.	2014	179 classifiers, 121 datasets: random forests win	PDF
Why do tree-based models still outperform deep learning on tabular data? — Grinsztajn et al.	2022	Empirical answer, still cited in 2026	arXiv:2207.08815
A Unified Approach to Interpreting Model Predictions (SHAP) — Lundberg & Lee	2017	The explainability tool you will be asked for in Module 12	arXiv:1705.07874
Data Cascades in High-Stakes AI — Sambasivan et al.	2021	Data-quality failures compound downstream; the empirical case for "data first"	PDF

Free books (legally downloadable, full text)

Book	Author	Link
An Introduction to Statistical Learning (ISLP, Python ed.)	James, Witten, Hastie, Tibshirani, Taylor	https://www.statlearning.com/
The Elements of Statistical Learning	Hastie, Tibshirani, Friedman	https://hastie.su.domains/ElemStatLearn/
Pattern Recognition and Machine Learning	Bishop (free since 2023)	https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/
Probabilistic Machine Learning: An Introduction	Kevin Murphy	https://probml.github.io/pml-book/book1.html
Mathematics for Machine Learning	Deisenroth, Faisal, Ong	https://mml-book.github.io/
Understanding Machine Learning: From Theory to Algorithms	Shalev-Shwartz & Ben-David	https://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning/

How to read an ML paper (30 minutes, 3 passes)

1. **Pass 1 (5 min)**: title, abstract, figures, conclusion. Ask: what problem, what's the claim?

2. **Pass 2 (15 min):** method section and experimental setup. Ask: what's the baseline, and is the comparison fair?
3. **Pass 3 (10 min):** limitations, ablations, and what you'd need to reproduce it. Ask: what would make this fail?

Keep a one-paragraph note per paper. In six months the note is worth more than the PDF.

Resources — ML Foundations

Courses (free)

Course	Who it's for	Link
Andrew Ng — Machine Learning Specialization (Coursera, audit free)	The canonical first course	https://www.coursera.org/specializations/machine-learning-introduction
fast.ai — Practical Deep Learning (Lesson 1–3 cover tabular ML)	Top-down, code-first	https://course.fast.ai/
Google — Machine Learning Crash Course	Fastest path to working vocabulary	https://developers.google.com/machine-learning/crash-course
CS229 Stanford — Machine Learning (notes + lectures)	The mathematical treatment	https://cs229.stanford.edu/
Kaggle Learn — Intro to ML / Intermediate ML	4 hours to a working model	https://www.kaggle.com/learn
StatQuest (Josh Starmer)	When an equation refuses to click	https://www.youtube.com/@statquest
MIT 6.036 / 6.390 Intro to ML	Rigorous, free courseware	https://openlearninglibrary.mit.edu/

Repositories worth cloning

Repo	What's inside
https://github.com/scikit-learn/scikit-learn	Read <code>sklearn/linear_model/_logistic.py</code> — production-grade versions of what you wrote
https://github.com/ageron/handson-ml3	Notebooks for <i>Hands-On ML with Scikit-Learn, Keras & TensorFlow</i> (3rd ed.)
https://github.com/rasbt/machine-learning-book	Raschka's <i>Machine Learning with PyTorch and Scikit-Learn</i>
https://github.com/microsoft/ML-For-Beginners	12-week structured curriculum from Microsoft
https://github.com/dmlc/xgboost	The library and its excellent docs/tutorials
https://github.com/eriklindernoren/ML-From-Scratch	Clean NumPy implementations of ~30 algorithms
https://github.com/probml/pyprobml	Code for Murphy's <i>Probabilistic Machine Learning</i>
https://github.com/shap/shap	Explainability you'll need in Module 12
https://github.com/evidentlyai/evidently	Drift & data-quality monitoring — pairs with §11 of the deep dive

Clone the starter set with: `bash ../_tools/clone_repos.sh 01-ml-foundations`

Datasets to practise on

Dataset	Why	Where
California Housing	clean regression, no leakage traps	<code>sklearn.datasets.fetch_california_housing()</code>
Adult / Census Income	categorical encoding + fairness discussion	<code>fetch_openml("adult", version=2)</code>
Titanic	the classic; great for feature engineering	https://www.kaggle.com/c/titanic
Credit Card Fraud	severe imbalance — use with §6 of the notes	https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
UCI ML Repository	600+ datasets	https://archive.ics.uci.edu/
Hugging Face Datasets	200k+ datasets, one-line loading	https://huggingface.co/datasets

Cheat sheets

- scikit-learn algorithm chooser: https://scikit-learn.org/stable/machine_learning_map.html
- scikit-learn model evaluation guide: https://scikit-learn.org/stable/modules/model_evaluation.html
- Also see `../../_shared/cheatsheets/`

Communities

- r/MachineLearning — <https://www.reddit.com/r/MachineLearning/>
- Papers with Code — <https://paperswithcode.com/>

- Hugging Face forums — <https://discuss.huggingface.co/>